

Special Notice

Industry Day Special Notice

POWER: Persistent Optical Wireless Energy Relay, Aerial
Demonstration

DARPA-SN-25-71

May 7, 2025



Defense Advanced Research Projects Agency

Tactical Technology Office (TTO)

675 North Randolph Street

Arlington, VA 22203-2114

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Defense Advanced Research Projects Agency (DARPA)
Tactical Technology Office (TTO)**

Posting Date: May 7, 2025

Special Notice (SN) DARPA-SN-25-71: POWER Aerial Demonstration Industry Day

Date: May 29, 2025

Location: DARPA Conference Center (DCC), 675 N Randolph Street, Arlington, VA 22203

Classification: Controlled Unclassified Information (CUI)

Registration Deadline: May 21, 2025 at 5:00 p.m. Eastern Time (ET)

Registration Website: <https://creative.spa.com/darpa/tto/power/id/2025/may/>

E-mail: EnergyWebDominance@darpa.mil

Technical POC: Dr. Paul Jaffe, TTO Program Manager

POWER PHASE 2 INDUSTRY DAY:

The Defense Advanced Research Projects Agency (DARPA) Tactical Technology Office (TTO) is sponsoring an Industry Day to provide information to potential proposers on the objectives of the second phase of the DARPA Persistent Optical Wireless Energy Relay (POWER) program, in advance of a planned Program Solicitation for Phase 2. The Industry Day will be held on **May 29, 2025, in the DCC, located at 675 N Randolph Street, Arlington, VA 22203** from **9:00 a.m. to 5:00 p.m. ET**. Advance registration is required.

The morning of the event will consist of government team briefings covering: POWER Program vision; Phase 1 accomplishments; Phase 2 objectives; and resources available to potential proposers.

The afternoon will begin with an interactive question and answer session regarding the planned Phase 2 solicitation, followed by company overview presentations.

PROGRAM OVERVIEW:

The goal of the POWER program is to design and demonstrate airborne optical energy relays that deliver 5 kW electrical power at a ground range of 200 km through 3 airborne relays. The relays will be suitable for integration on a host aircraft, have an aperture of less than 1 m, and be capable of harvesting 1 kW electrical power from the relayed beam for own ship use. The principal technical challenges are to develop and demonstrate a payload capable of redirecting optical energy transmissions, correcting wavefronts to maximize beam quality at each relay, and selectively harvesting energy for platform power.

POWER Phase 2 will challenge performers to rapidly mature and demonstrate a simulated ground-air-ground relay link in a representative environment at the White Sands Missile Range High Energy Laser Test Facility (HELSTF). This effort will exercise the interfaces and overall system concept of operations, building confidence that full-scale designs are capable of meeting POWER vision metrics described above. Multiple performers are expected to use largely commercial-off-the-shelf components and heavyweight custom hardware to achieve this demonstration.

REGISTRATION INFORMATION:

Participants must register for the event through the registration website:

<https://creative.spa.com/darpa/tto/power/id/2025/may/>

Due to space limitations of the facility, attendance will be limited to the first 100 registrants, and no more than four (4) representatives per organization.

Upon entry to the event, all attendees will be required to present Government-issued photo identification that is compliant with the REAL ID Act of 2005, and an acceptable proof of U.S. citizenship or permanent resident status (Green Card holders). Examples may include:

- State-issued driver's license, or,
- U.S. passport, or,
- U.S. Permanent Resident card, or,
- Copy of birth certificate.

See <https://www.darpa.mil/policy/visitor-information> for additional information on identification requirements for entry.

NOTE: Before individuals transmit Personally Identifiable Information (PII) or other sensitive data to DARPA, it is recommended that they apply appropriate physical, administrative, and technical protocols to safeguard their privacy. DARPA will secure all received PII and sensitive personal data in a controlled environment and protect it from unauthorized access, use, or disclosure.

SECURITY INFORMATION:

The event will be conducted at the Controlled Unclassified Information (CUI) level and is open only to U.S. persons. All attendees accessing CUI agree to protect CUI in accordance with DoD Instruction 5200.48 CONTROLLED UNCLASSIFIED INFORMATION (CUI) and NIST Special Publication 800-171 Protecting Controlled Unclassified Information in Nonfederal Systems and Organizations. All attendees will be required to present government issued photo identification when requested by the security POC.

For instructions on sending visit requests and security clearances to DARPA, please visit <http://www.darpa.mil/policy/visitor-information>.

No cell phones, electronic devices, recording, or photography will be permitted during the event at DARPA. Storage is available for small electronics.

There will be no exceptions to this requirement.

DARPA will not post any attendee information presented at the event on the Agency's public website.

TEAMING OPPORTUNITIES:

DARPA hosts Industry Days to promote teaming arrangements between researchers; provide potential performers with information on whether and how they might respond to the Government's research and development solicitations; and to increase efficiency in proposal preparation and evaluation.

Attendees who would like to present company overviews, discuss technology expertise, and/or discuss teaming opportunities may request to do so in scheduled presentations at the event. Presentations may contain up to 5 slides and are limited to 5 minutes of presentation time and up to 5 additional minutes of questions and answers. Attendees should contact EnergyWebDominance@darpa.mil, as soon as possible to request a presentation time at the event; requests must be received no later than the registration cutoff date. Requests will be considered on a first-come, first-served basis. 20 available slots are anticipated. If the slots are filled, requesters will be added to a waitlist in case slots become available. Proposed presentations must be relevant to the POWER program.

All presentation materials must be approved in advance by both the organization that funded the research and the DARPA Program Manager. The DARPA Program Manager will screen the proposed material for controlled unclassified information (CUI) and approve presentations in advance of the event. It is the presenter's responsibility to ensure that all material they plan to present at the event has been approved by the funding organization in advance. An electronic copy of all material to be presented must be provided to EnergyWebDominance@darpa.mil for review by May 22, 2025.

A list of registrant contact information will be made available to attendees following the event to facilitate interchange among interested proposers. **If you would NOT like to be included on this list, please notify us when registering.**

ADDITIONAL INFORMATION:

A draft program solicitation will be posted to this Special Notice no later than May 21, 2025.

Attendance at the POWER Phase 2 Industry Day is voluntary and is not required to propose to subsequent Program Solicitations (if any) on this topic. The Industry Day does not constitute a formal solicitation for proposals or abstracts. This announcement

is issued solely for information and program planning purposes and is not a Request for Information (RFI). Since this is not an RFI, no submissions against this notice will be accepted by the Government. DARPA will not provide reimbursement for costs incurred to participate in this event.